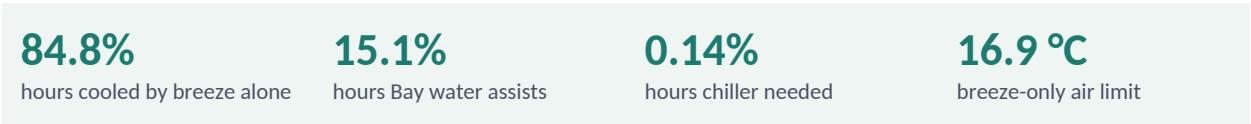


From Bay Breeze to Server Supply Air: A Year-Round Heat Exchanger Model for a Hybrid-Cooled Pier Data Center

Vishnu Sai Sharan Ankathi • Independent research • September 2026 • Python (ε-NTU model), 3 years of hourly data



1. Summary

This analysis tests the central claim of the project: that the San Francisco Bay's cool marine-layer air can do most of the work of cooling a data center, with pumped Bay water held in reserve. It models the full cooling chain hour by hour for 2022–2024. Hot room air passes first through an air-to-air heat exchanger cooled by outside air, then, only if needed, through an air-to-seawater coil, with a chiller trimming anything left.

Result: for a 270 kW room with 21 °C supply air, the breeze alone does the job 84.8% of the year. Bay water covers another 15.1%, and the chiller is needed for just 0.14% of hours (peak about 17 kW of cooling). Even on the hottest day in the record, when outside air reached ~38 °C, Bay water held the supply at 21 °C.

2. Why a Heat Exchanger Chain

Bay air is cool but humid, averaging 78.7% relative humidity. Blowing it straight into the server room meets ASHRAE's recommended envelope only 3.9% of the time (40.0% of the wider A2 allowable envelope). The design therefore uses indirect cooling: outside air and room air stay in separate streams and only exchange heat across a heat exchanger wall. Humidity stops mattering; only temperature does.

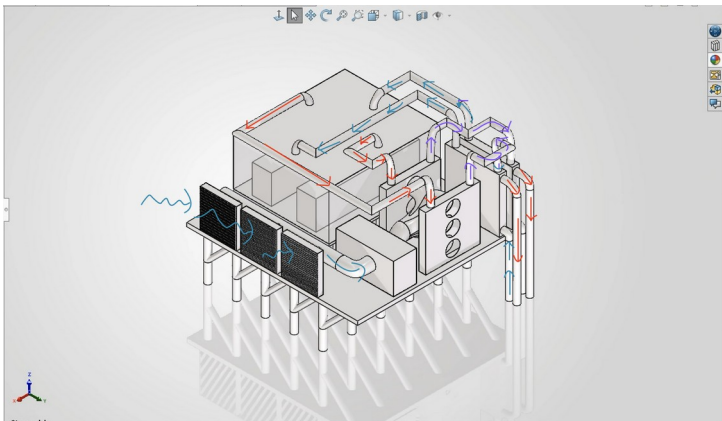


Figure 1. Concept CAD: filter banks feed outside air to the heat exchangers; bay water is pumped to a second-stage coil; the sealed server room never sees outside air.

Stage	What it does	When it runs
1. Breeze (air-to-air HX)	Outside marine air cools the room's	Always first; fans modulate so supply

Stage	What it does	When it runs
	return air	never drops below setpoint
2. Bay water (air-to-seawater coil)	Pumped seawater cools the air further	Only if Stage 1 leaves the air above setpoint
3. Chiller	Mechanical cooling trims the remainder	Only if both stages together fall short

3. Method

3.1 Data

- Hourly air temperature, relative humidity and dew point, 2022–2024 (26,304 hours), from the Open-Meteo historical archive (ERA5 reanalysis) for a San Francisco Bay coastal point.
- Hourly water temperature from NOAA CO-OPS station 9414290 (San Francisco). Measured data runs to February 2024; later hours are filled with month-by-hour averages of the measured record.

3.2 Design inputs

Input	Value	Basis
Room heat load	270 kW (2 paths × 135 kW)	18 racks × 15 kW
Room airflow	10.06 m ³ /s per path	Sized for ~11 °C rise across racks
Return (hot-aisle) air	33.25 °C	Room CFD model of the same layout
Supply setpoint	21 °C (24 °C tested)	ASHRAE recommended 18–27 °C
Stage 1 HX	Cross-flow, unmixed; design $\epsilon = 0.75$	Typical air-to-air plate/coil unit
Stage 2 coil	Counterflow; design $\epsilon = 0.80$; 3.25 kg/s seawater per path	Air-to-water coil
Chiller	COP 5	Typical water-cooled chiller

3.3 Model

Each heat exchanger is modeled with the effectiveness-NTU method. Stage 1's conductance (UA) is fixed by its design effectiveness at equal outside and room airflow; raising outside airflow then increases its effectiveness through the NTU relations. For each hour the model: (1) cools the return air with outside air, never below setpoint and bypassing Stage 1 if outside air is hotter than the return air; (2) adds the seawater coil only if needed; (3) assigns any remaining load to the chiller.

$$\text{Supply after Stage 1} = T_{\text{return}} - \epsilon_1 \times (T_{\text{return}} - T_{\text{outside}}); \text{ after Stage 2} = T_1 - \epsilon_2 \times (T_1 - T_{\text{water}})$$

4. Results

4.1 The breeze-only limit

With $\epsilon_1 = 0.75$ and a 33.25 °C return, outside air at or below about 16.9 °C is enough to deliver 21 °C supply air on its own. Through most of the year Bay air sits below that line; in late summer and early fall it crosses it more often, and Bay water, which stays around 15–19 °C, takes over.

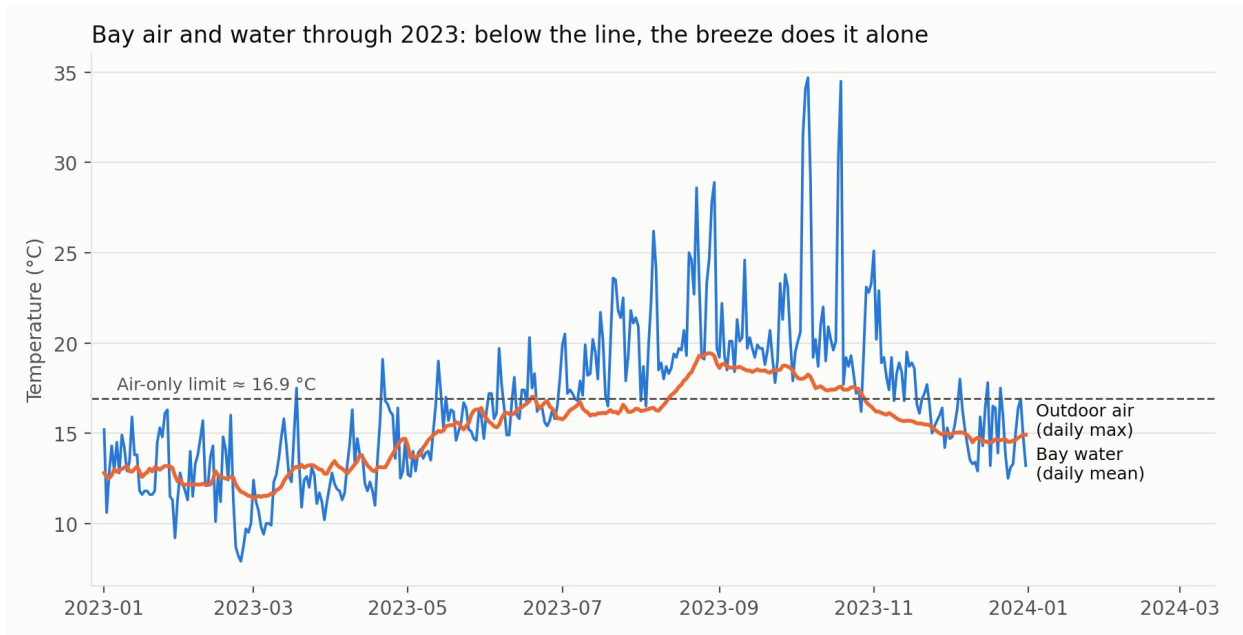


Figure 2. Daily maximum outdoor air and daily mean Bay water temperature, 2023, against the breeze-only limit.

4.2 Which stage does the work

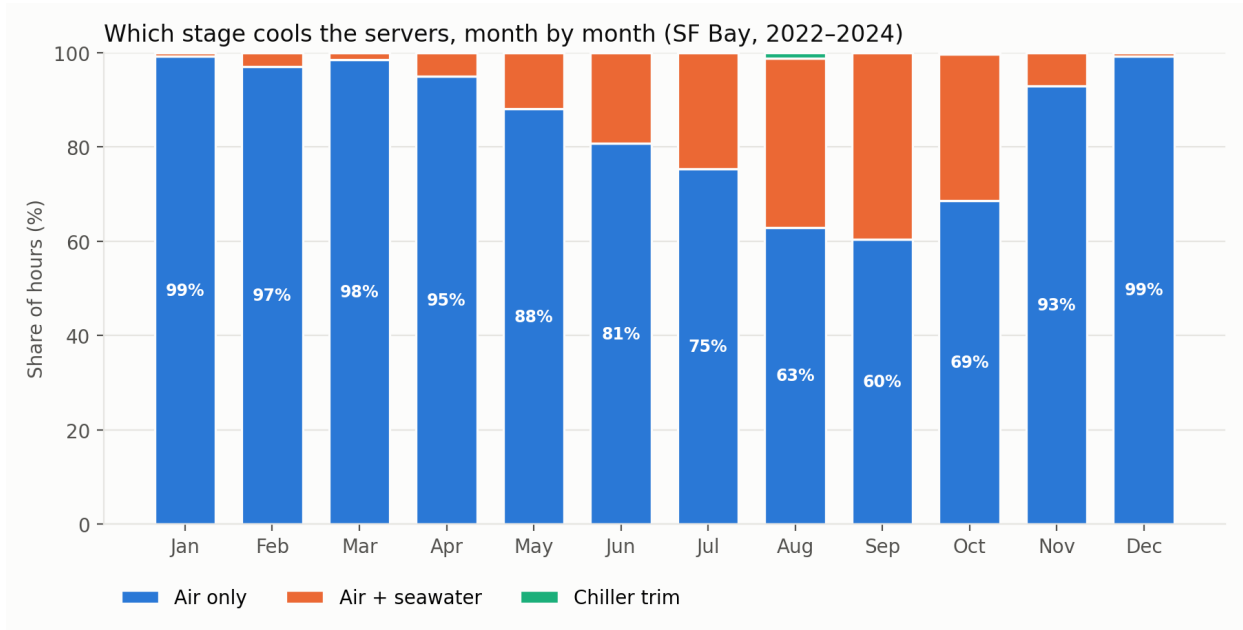


Figure 3. Share of hours by cooling stage, by month, 2022-2024.

Result (21 °C supply, $\epsilon_1 = 0.75$)	Value
Breeze only	84.8% of hours
Breeze + Bay water	15.1% of hours (= seawater pump runtime)
Chiller trim	0.14% of hours, peak ~17 kW thermal, ~0.01 MWh electric per year
Seawater pump energy	~4.5 MWh electric per year
Hardest month	September: breeze alone 60% of hours

4.3 The hottest day

On 6 September 2022, outside air peaked near 38 °C, hotter than the room's return air. The breeze stage was bypassed at midday, and the seawater coil alone brought the air down to 21 °C. The chiller was not needed.

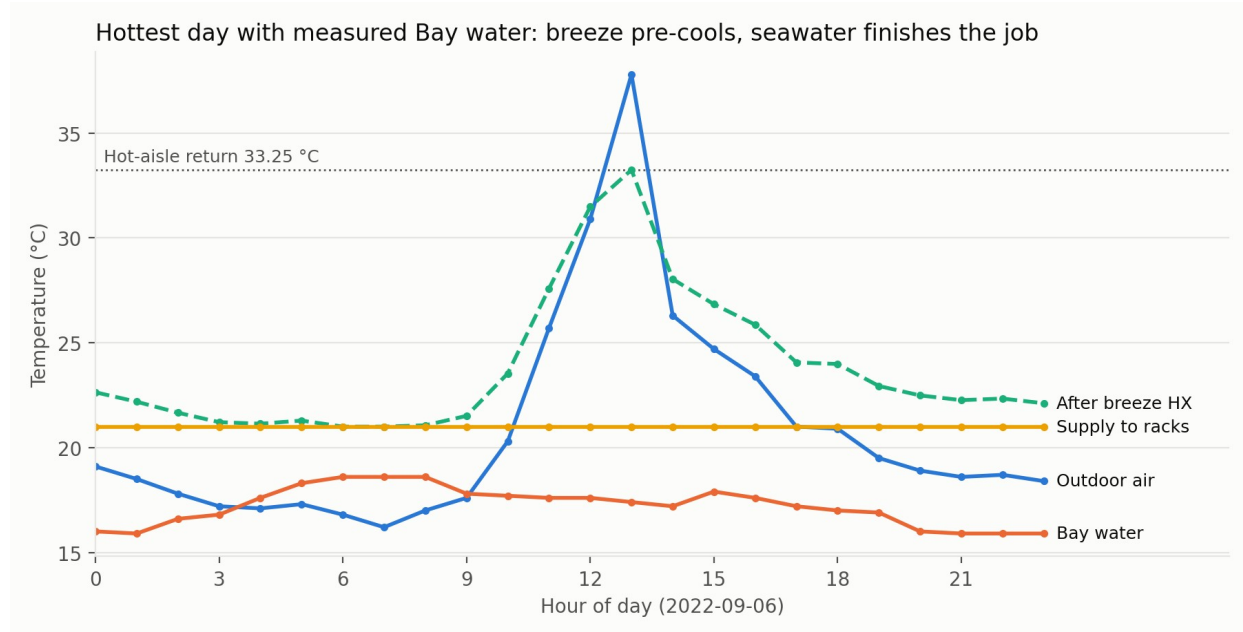


Figure 4. Temperature chain on the hottest day with measured Bay water: outdoor air, Bay water, air after the breeze stage, and final supply to the racks.

4.4 Sensitivity

Heat exchanger effectiveness is the biggest assumption. The table shows breeze-only hours at 21 °C supply across realistic values, with outside airflow equal to or 1.5× the room airflow.

Stage 1 design ϵ	Outside/room airflow 1.0	Outside/room airflow 1.5
0.65	63%	83%
0.75	85%	93%
0.85	93%	96%

Raising the supply setpoint to 24 °C, still inside ASHRAE's recommended range, pushes breeze-only hours to 87–99% and removes the chiller entirely in every case tested. This assumes the return air rises with the supply (36.25 °C at 24 °C supply); a CFD run at 24 °C will confirm it.

5. Findings

- **The breeze is the primary cooler.** It covers about 85% of the year at 21 °C supply, and more with a better exchanger, more outside airflow or a warmer setpoint.
- **Bay water is a true backup, not the main system.** The pump runs about 15% of hours, versus continuously in existing water-cooled designs, which reduces seawater intake, discharge and corrosion exposure.

- **The chiller is almost idle**, but still needed for redundancy and extreme days.
- **Late summer and early fall are the design-critical season**, when both air and water are warmest.
- **A correction to the earlier screen:** a simpler approach-temperature screen gave ~90% air-only hours. This ϵ -NTU model is more realistic and gives 84.8%.

6. Limitations

- One weather grid point and one tide station, not an on-site survey; three years of data, not a 30-year climate normal.
- Heat exchanger effectiveness values are design assumptions, not vendor data; the sensitivity table brackets them.
- Fan power and pressure drops through filters, ducts and coils are not yet included.
- Steady hourly model: no pump start-up time, switching hysteresis or thermal lag.
- Seawater fouling and salt-aerosol build-up on the outdoor side would lower effectiveness over time.

7. Next Steps

- Fan power from filter, duct and coil pressure drops, for a full PUE estimate.
- Hour-by-hour control simulation with pump hysteresis and a seawater runtime budget.
- Warm-supply stress case in the room CFD model (24–25 °C supply).
- Vendor data for real air-to-air and seawater heat exchangers to replace assumed effectiveness.

Code and data: the bay-breeze-cooling repository (Python). Data: Open-Meteo (CC BY 4.0) and NOAA CO-OPS (public domain).

Research idea, design decisions and modeling direction by the author. AI tools were used to help write code, run calculations and draft this report.